

The `latexasm` package*

Sergio Martinez-Losa

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1 Introduction

The `latexasm` package implements a virtual machine (VM) within `LATEX` using `expl3` that executes assembly-like code. This allows you to write programs with registers, arithmetic operations, control flow, and string manipulation directly in your `LATEX` documents.

2 Usage

2.1 Basic Example

```
\usepackage{latexasm}

\begin{latexasm}
  \Instr{MOV}[R0][5]
  \Label{LOOP}
  \Instr{PRINT}[R0]
  \Instr{SUB}[R0][1]
  \Instr{CMP}[R0][0]
  \Instr{JNZ}[LOOP]
  \Instr{HALT}
\end{latexasm}
```

2.2 Registers

The VM provides 8 general-purpose registers: `R0`, `R1`, `R2`, `R3`, `R4`, `R5`, `R6`, and `R7`.

2.3 Instructions

`MOV dest, src` Move value from `src` to `dest`

`ADD dest, src` Add `src` to `dest`

*This document corresponds to `latexasm` ?, dated ?.

SUB `dest`, `src` Subtract `src` from `dest`

MUL `dest`, `src` Multiply `dest` by `src`

DIV `dest`, `src` Divide `dest` by `src`

MOD `dest`, `src` Modulo `dest` by `src`

CMP `a`, `b` Compare `a` and `b`, set flags

JMP `label` Unconditional jump to `label`

JZ `label` Jump to `label` if zero flag is set

JNZ `label` Jump to `label` if zero flag is not set

PRINT `src` Print numeric value to PDF

CHR `src` Print ASCII character to PDF

DATA `label`, `value` Define named string or constant

STRLEN `dest`, `label` Get length of string

GETCH `dest`, `label`, `index` Get ASCII value of character

HALT Stop execution

2.4 Examples

See the included `example.tex` file for complete examples including: countdown loops, Hello World, atoi implementation, and comprehensive instruction usage.

3 Implementation

```

1 \RequirePackage{xparse}
2
3 % --- Registers ---
4 \prop_new:N \g_latexasm_registers_prop
5 \int_new:N \g_latexasm_pc_int
6
7 % --- Flags ---
8 \bool_new:N \g_latexasm_z_flag_bool
9 \bool_new:N \g_latexasm_n_flag_bool
10 \bool_new:N \g_latexasm_halt_bool
11
12 % --- Memory and Data ---
13 \prop_new:N \g_latexasm_memory_prop
14 \prop_new:N \g_latexasm_labels_prop
15 \seq_new:N \g_latexasm_code_seq
16

```

```

17 % --- Internal Variables ---
18 \tl_new:N \l_latexasm_tmp_tl
19 \int_new:N \l_latexasm_tmp_int
20
21 % --- API ---
22
23 \NewDocumentEnvironment{latexasm}{}
24 {
25   \seq_gclear:N \g_latexasm_code_seq
26   \prop_gclear:N \g_latexasm_labels_prop
27   \prop_gclear:N \g_latexasm_registers_prop
28   % Initialize registers R0-R7
29   \clist_map_inline:nn { R0, R1, R2, R3, R4, R5, R6, R7 }
30     { \prop_gput:Nnn \g_latexasm_registers_prop { ##1 } { 0 } }
31   \int_gset:Nn \g_latexasm_pc_int { 0 }
32   \bool_gset_false:N \g_latexasm_halt_bool
33 }
34 {
35   \latexasm_run:
36 }
37
38 \cs_new_protected:Npn \latexasm_run:
39 {
40   \bool_until_do:nn { \g_latexasm_halt_bool }
41   {
42     \latexasm_step:
43   }
44 }
45
46 \cs_new_protected:Npn \latexasm_step:
47 {
48   \int_gincr:N \g_latexasm_pc_int
49   \l_latexasm_tmp_int = \int_use:N \g_latexasm_pc_int
50   \tl_set:Nx \l_latexasm_tmp_tl { \seq_item:Nn \g_latexasm_code_seq { \l_latexasm_tmp_int } }
51
52   \tl_if_empty:NTF \l_latexasm_tmp_tl
53     { \bool_gset_true:N \g_latexasm_halt_bool }
54     { \latexasm_dispatch:V \l_latexasm_tmp_tl }
55 }
56
57 \cs_new_protected:Npn \latexasm_dispatch:n #1
58 {
59   % Very simple dispatcher for now
60   % Format: \cmd {arg1} {arg2} ...
61   #1
62 }
63 \cs_generate_variant:Nn \latexasm_dispatch:n { V }
64
65 % --- Instructions ---
66

```

```

67 \cs_new_protected:Npn \L_HALT
68   { \bool_gset_true:N \g_latexasm_halt_bool }
69
70 \cs_new_protected:Npn \L_MOV #1 #2
71   {
72     \latexasm_set_val:nn { #1 } { \latexasm_get_val:n { #2 } }
73   }
74
75 \cs_new_protected:Npn \L_ADD #1 #2
76   {
77     \int_set:Nn \l_latexasm_tmp_int { \latexasm_get_val:n { #1 } + \latexasm_get_val:n { #2 } }
78     \latexasm_set_val:nn { #1 } { \int_use:N \l_latexasm_tmp_int }
79     \latexasm_update_flags:n { \l_latexasm_tmp_int }
80   }
81
82 \cs_new_protected:Npn \L_SUB #1 #2
83   {
84     \int_set:Nn \l_latexasm_tmp_int { \latexasm_get_val:n { #1 } - \latexasm_get_val:n { #2 } }
85     \latexasm_set_val:nn { #1 } { \int_use:N \l_latexasm_tmp_int }
86     \latexasm_update_flags:n { \l_latexasm_tmp_int }
87   }
88
89 \cs_new_protected:Npn \L_CMP #1 #2
90   {
91     \int_set:Nn \l_latexasm_tmp_int { \latexasm_get_val:n { #1 } - \latexasm_get_val:n { #2 } }
92     \latexasm_update_flags:n { \l_latexasm_tmp_int }
93   }
94
95 \cs_new_protected:Npn \L_MUL #1 #2
96   {
97     \int_set:Nn \l_latexasm_tmp_int { \latexasm_get_val:n { #1 } * \latexasm_get_val:n { #2 } }
98     \latexasm_set_val:nn { #1 } { \int_use:N \l_latexasm_tmp_int }
99     \latexasm_update_flags:n { \l_latexasm_tmp_int }
100  }
101
102 \cs_new_protected:Npn \L_DIV #1 #2
103   {
104     \int_set:Nn \l_latexasm_tmp_int { \latexasm_get_val:n { #1 } / \latexasm_get_val:n { #2 } }
105     \latexasm_set_val:nn { #1 } { \int_use:N \l_latexasm_tmp_int }
106     \latexasm_update_flags:n { \l_latexasm_tmp_int }
107   }
108
109 \cs_new_protected:Npn \L_MOD #1 #2
110   {
111     \int_set:Nn \l_latexasm_tmp_int { \int_mod:nn { \latexasm_get_val:n { #1 } } { \latexasm_ge
112     \latexasm_set_val:nn { #1 } { \int_use:N \l_latexasm_tmp_int }
113     \latexasm_update_flags:n { \l_latexasm_tmp_int }
114   }
115
116 \cs_new_protected:Npn \L_JMP #1

```

```

117 {
118   \prop_get:NnNTF \g_latexasm_labels_prop { #1 } \l_latexasm_tmp_tl
119   { \int_gset:Nn \g_latexasm_pc_int { \l_latexasm_tmp_tl - 1 } }
120   { \msg_error:nnn { latexasm } { unknown-label } { #1 } }
121 }
122
123 \cs_new_protected:Npn \L_JZ #1
124 {
125   \bool_if:NT \g_latexasm_z_flag_bool { \L_JMP { #1 } }
126 }
127
128 \cs_new_protected:Npn \L_JNZ #1
129 {
130   \bool_if:NF \g_latexasm_z_flag_bool { \L_JMP { #1 } }
131 }
132
133 \cs_new_protected:Npn \L_PRINT #1
134 {
135   \mode_leave_vertical:
136   \latexasm_get_val:n { #1 } \space
137 }
138
139 \cs_new_protected:Npn \L_CHR #1
140 {
141   \mode_leave_vertical:
142   \char\latexasm_get_val:n { #1 }
143 }
144
145 \cs_new_protected:Npn \L_DATA #1 #2
146 {
147   \prop_gput:Nnn \g_latexasm_memory_prop { #1 } { #2 }
148 }
149
150 \cs_new_protected:Npn \L_STRLEN #1 #2
151 {
152   \prop_get:NnNTF \g_latexasm_memory_prop { #2 } \l_latexasm_tmp_tl
153   { \latexasm_set_val:nn { #1 } { \str_count:N \l_latexasm_tmp_tl } }
154   { \latexasm_set_val:nn { #1 } { 0 } }
155 }
156
157 \cs_new_protected:Npn \L_GETCH #1 #2 #3
158 {
159   \prop_get:NnNTF \g_latexasm_memory_prop { #2 } \l_latexasm_tmp_tl
160   {
161     \tl_set:Nx \l_latexasm_tmp_tl { \str_item:Nn \l_latexasm_tmp_tl { \latexasm_get_val:n { #1 } } }
162     \latexasm_set_val:nn { #1 } { \exp_after:wN ' \l_latexasm_tmp_tl }
163   }
164   { \latexasm_set_val:nn { #1 } { 0 } }
165 }
166

```

```

167 % --- Helpers ---
168
169 \cs_new:Npn \latexasm_get_val:n #1
170 {
171   \prop_if_in:NnTF \g_latexasm_registers_prop { #1 }
172     { \prop_item:Nn \g_latexasm_registers_prop { #1 } }
173     { #1 }
174 }
175
176 \cs_new_protected:Npn \latexasm_set_val:nn #1 #2
177 {
178   \prop_gput:Nnx \g_latexasm_registers_prop { #1 } { \int_eval:n { #2 } }
179 }
180
181 \cs_new_protected:Npn \latexasm_update_flags:n #1
182 {
183   \int_compare:nNnTF { #1 } = { 0 }
184     { \bool_gset_true:N \g_latexasm_z_flag_bool }
185     { \bool_gset_false:N \g_latexasm_z_flag_bool }
186
187   \int_compare:nNnTF { #1 } < { 0 }
188     { \bool_gset_true:N \g_latexasm_n_flag_bool }
189     { \bool_gset_false:N \g_latexasm_n_flag_bool }
190 }
191
192 % --- Syntax Sugar ---
193
194 \NewDocumentCommand{\Instr}{ m o o o }
195 {
196   \IfNoValueTF { #2 }
197     { \seq_gput_right:Nn \g_latexasm_code_seq { \use:c { L_#1 } } }
198     {
199       \IfNoValueTF { #3 }
200       { \seq_gput_right:Nn \g_latexasm_code_seq { \use:c { L_#1 } { #2 } } }
201       {
202         \IfNoValueTF { #4 }
203         { \seq_gput_right:Nn \g_latexasm_code_seq { \use:c { L_#1 } { #2 } { #3 } } }
204         { \seq_gput_right:Nn \g_latexasm_code_seq { \use:c { L_#1 } { #2 } { #3 } { #4 } } }
205       }
206     }
207 }
208
209 \NewDocumentCommand{\Label}{ m }
210 {
211   \prop_gput:Nnx \g_latexasm_labels_prop { #1 } { \int_eval:n { \seq_count:N \g_latexasm_code
212 }
213
214 \msg_new:nnnn { latexasm } { unknown-label }
215 { Unknown ~ label ~ '1'. }
216 { Check ~ your ~ label ~ definitions. }

```

